

Football Trajectory Prediction with Conformal Regions

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Introduction

Predicting where players move next is central to football analytics. We adapt pedestrian trajectory forecasting to football and quantify its uncertainty with conformal prediction. We train a dynamic social decoder on roughly one hundred plays to forecast all outfield players five seconds ahead, with conformal prediction for finite-sample trajectory regions. Broadcast graphics, dashboards, and scouting tools need accurate forecasts with coverage guarantees.

Methods

Data. Open Bundesliga tracking data: seven matches, ten teams, 164 open-play shot plays, 3,253 player trajectories. Each play provides 25 observed seconds plus a 5-second horizon, orientation-normalized; plays split 60/15/15/10 into train/validation/calibration/test prevents frame leakage across sets.

Architecture. The model adapts pedestrian-trajectory predictors (Social-BiGAT/STGAT, Kosaraju et al., 2019; Huang et al., 2019) to deterministic forecasting: a spatiotemporal encoder (per-frame multi-head attention, then LSTM) and an autoregressive decoder re-computing graph attention over all players’ states at each future step, predicting residual displacement against a constant-velocity prior; lateral mirroring doubles training data.

Conformal regions. Around the point predictor, for play k , player i , horizon t , define the error $e_{i,t} = \|p_i(t) - \hat{p}_i(t)\|_2$ and trajectory-level nonconformity score $R_{k,i} = \max_t e_{i,t} / \hat{s}_t(x_i)$. The band shape $\hat{s}_t$ is estimated from out-of-fold training residuals; the adaptive scale $\hat{s}_t(x_i) = \exp\{\hat{g}(x_i, t)\}$ uses ridge regression on the frozen latent state $h_i(T_{\text{obs}})$ plus horizon t . Calibration yields the order statistic $\hat{q} = R_{(\lceil (n+1)(1-\alpha) \rceil)}$, $\alpha = 0.10$, giving regions $C_t = \{p : \|p - \hat{p}(t)\|_2 \leq \hat{q} \hat{s}_t(x_i)\}$. Trajectories within a play being dependent, the calibration unit is the (play, player) pair; exact finite-sample control follows from conformal risk control (Angelopoulos et al. 2024) and cross-conformal (Vovk 2015; Barber et al. 2021) with five play folds. Region efficiency is measured by mean area πr^2 .

Results

On validation, the final model attains average displacement error (ADE) 4.06 m and final displacement error (FDE) 6.55 m; on the held-out test set, ADE 3.86 m and FDE 6.28 m, improving both attacking (3.78/6.25 m) and defending (3.93/6.31 m) players. At $\alpha = 0.10$, the split-conformal pipeline reaches 88.9% empirical simultaneous coverage (target 90%) with $\hat{q} = 3.42$; with only 24 calibration plays, $\hat{q}$ is noisy. Figure 1 shows a best-case attacking example with tight 90% conformal regions at $t = 1, 3$, and 5 s.

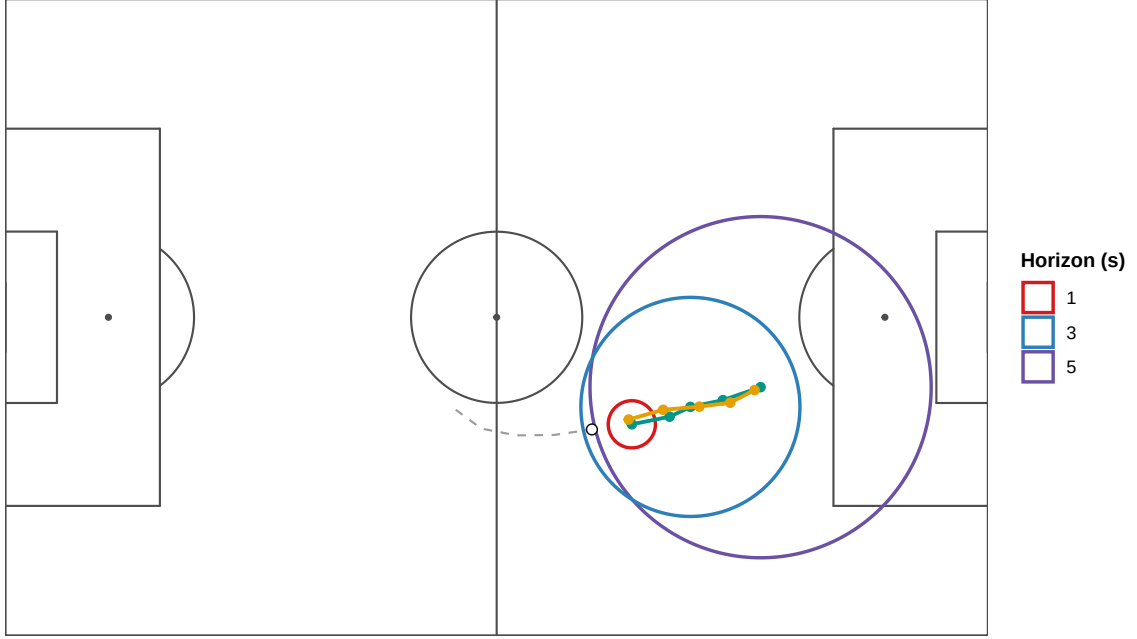


Figure 1: Figure 1: Five-second forecast for one attacking player. Green: predicted; orange: actual; grey dashed: last 5 s observed; rings: 90% conformal regions at $t = 1, 3, 5$ s (ADE 0.82 m).

Figure 2 shows the same regions for three attacking players.

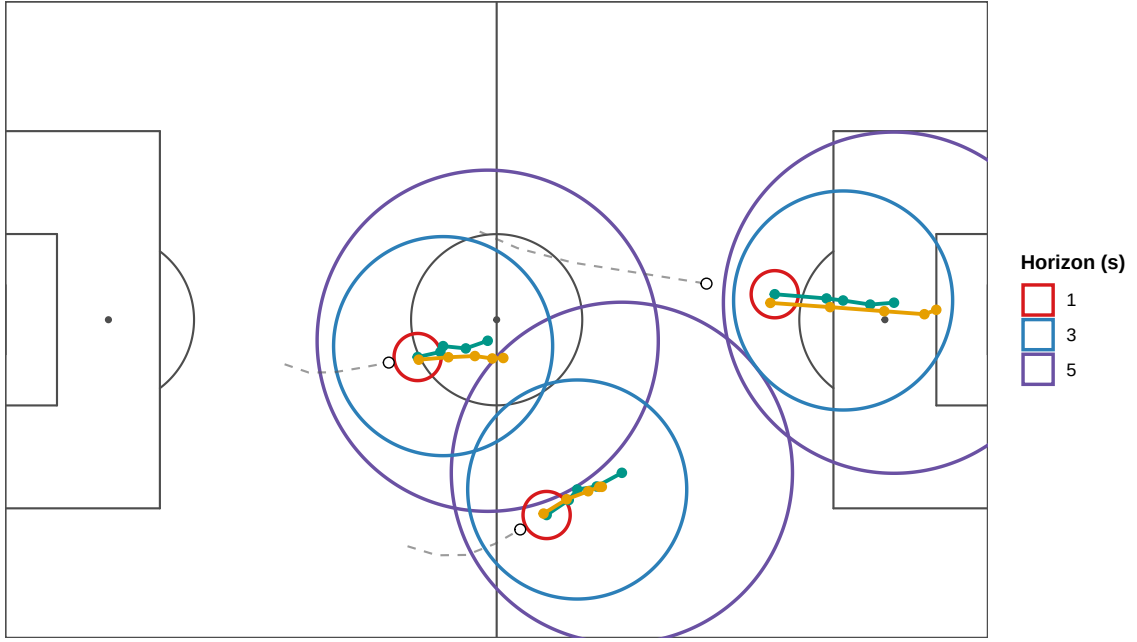


Figure 2: Figure 2: Five-second forecasts for three attacking players in the same play. Green: predicted; orange: actual; grey dashed: last 5 s observed; rings: 90% conformal regions at $t = 1, 3, 5$ s.

Conclusion

We adapt multi-agent trajectory prediction to football under limited data and equip forecasts with distribution-free, play-level uncertainty. The social decoder delivers accurate five-second forecasts; conformal refinements turn them into principled regions. Calibrated bands around player trajectories support broadcast, scouting, and decision tools—a template for small-sample, interdependent-trajectory uncertainty quantification.